Certified Quantum Tumor Generation: Full Pipeline and Literature

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Summary

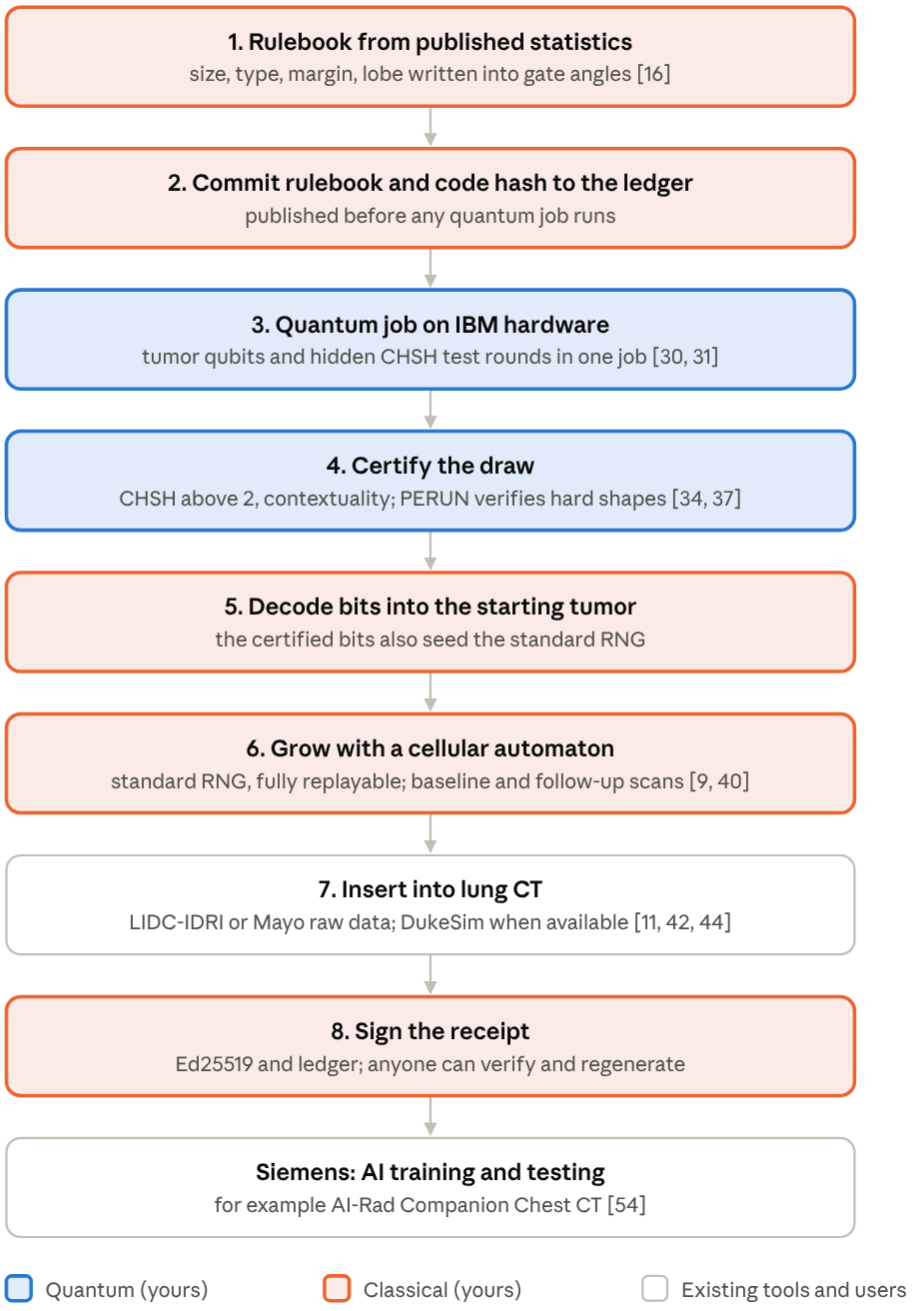
The thesis builds virtual lung tumors whose starting state is drawn by a quantum computer, certified as unchosen, grown classically, inserted into real CT scans and signed so anyone can verify them. Every stage has published precedent; no published work combines them, which is the contribution.

The pipeline in eight stages:

1. **Procedural generation of lesions** — established in virtual imaging trials and synthetic-tumor research [1–11].
2. **Quantum procedural generation** — quantum circuits already generate seeds for maps and images [12–15].
3. **Mapping the tumor rulebook to a circuit** — dependent clinical variables encoded as a quantum Bayesian network, never trained (no QML) [16–27].
4. **Quantum randomness as the seed** — does not improve realism [28, 29]; it makes the draw unchosen by anyone.
5. **Certification** — Bell (CHSH) tests in the same job prove the bits are quantum [30–39].
6. **Classical growth** — a cellular automaton driven by a standard RNG seeded from the certified bits [9, 40, 41].
7. **Insertion into CT** — validated image-domain and projection-domain methods [11, 42–47].
8. **Provenance and audit** — a signed receipt per tumor makes every image replayable and checkable [2, 3].

Sections 9–11 cover why this is useful, why quantum and its honest limits, and what is new.

Only the starting tumor is quantum; everything after it is replayable



certified tumor pipeline · 8 stages, then use

The diagram numbers steps in the order they run, while the sections below follow the literature. Quantum hardware touches only steps 3–4; every later step is classical, deterministic and replayable from the signed receipt.

1. Procedural generation of lesions: where it starts

Generating virtual patients and lesions by rules is an established method in medical imaging, used for AI training and even regulatory evidence.

Virtual imaging trials. A virtual imaging trial replaces patients with computational phantoms, the scanner with a simulator and the reader with a model observer [1]. The AAPM task group TG387 is standardising the field around credibility, reproducibility and accessibility [2]. FDA researchers review how digital cohorts are sampled and warn of study bias from selecting particular patient distributions [3].

Regulatory precedent. The FDA VICTRE trial imaged 2,986 computer-generated patients with inserted lesions and reproduced the result of a real clinical trial; the authors conclude in-silico trials can serve as regulatory evidence [4].

Procedural phantoms. Penn and Lund build breast phantoms algorithmically from geometric primitives and rules [5]. Duke's XCAT 3.0 library holds over 2,500 phantoms derived from CT [6]. Duke's SYN-LUNGS placed 512 procedurally generated lung nodules into 174 digital twins and simulated 1,044 CT scans for AI training [7].

Synthetic tumors for AI. Johns Hopkins showed an AI trained only on rule-based synthetic liver tumors segments real tumors about as well as one trained on real tumors [8]. Their Pixel2Cancer grows tumors with cellular-automaton rules (growth, invasion, death); three radiologists judged them realistic [9]. A review summarises rule-based and learned approaches [10]. For lungs specifically, Ambrosini and O'Dell published a rule-based nodule insertion recipe [11].

2. Quantum procedural generation

Quantum circuits have already been used to generate the starting structure of procedural content on real IBM hardware, with classical code completing it. That split is exactly this thesis's design.

- **Procedural generation using quantum computation** (Wootton, IBM) encodes an image or height map into qubits, applies fixed rotations and measures a varied version ("quantum blur") [12].
- **A quantum procedure for map generation** (Wootton, IBM) generates game terrain on IBM hardware: the quantum computer supplies the seed structure and classical code builds the map [13].
- **Investigating the usefulness of quantum blur** evaluates when the quantum method adds anything over a classical one, a model for the honest evaluation this thesis needs [14].
- **Quantum Wave Function Collapse** (Fraunhofer) maps a constraint-based tile generator onto qubits, relevant if neighbouring tumor features must obey local rules [15].

None of these work in medicine or certify their randomness; both gaps are filled here.

3. Mapping the tumor rulebook to a quantum circuit

The tumor's clinical properties are encoded as a quantum Bayesian network: published frequencies become gate angles, dependencies become controlled rotations, and one measurement gives one tumor. Nothing is trained, so this is not quantum machine learning.

The encoding method. Low, Yoder and Chuang show how any Bayesian network becomes a circuit of controlled rotations whose measurement samples the network [16]. Here the network is size → type → margin → location, with probabilities from published screening statistics.

Register	Qubits	Values	Conditioned on
Size	2	four size bands	—
Type	2	solid, part-solid, ground-glass, calcified	size
Margin	2	smooth, lobulated, spiculated	type
Location	3	five lung lobes	—
Shape seed	3–5	raw bits for surface detail	—
CHSH pair	2	certificate, same job	—

Precedents for mapping living behaviour to gates. Quantum artificial life encodes self-replication, mutation, interaction and death into gates [17] and ran on IBM hardware, with entanglement inherited across generations [18]. Quantum versions of Conway's Game of Life map cellular rules to circuits [19, 20]. IBM Zurich's protein-folding work is a full template for biology → qubit encoding → resource count → hardware run [21].

Where quantum helps biology, and where it is hype. Two reviews map the field and its limits, especially data loading and noise [22, 23].

Can quantum simplify a biological process? Yes, in one proven sense: a quantum simulator of a step-by-step random process can need less memory than any classical one [24], confirmed experimentally [25, 26]. This is a memory advantage, not speed or realism. Classical tensor-network methods already compress biological stochastic models with low correlations [27], so a quantum advantage needs high correlations. Tumor growth as a candidate is future work, not a claim.

4. Quantum randomness for better procedural generation

Quantum randomness does not make tumors more realistic; it makes the choice of tumors provably unchosen. That is the claim to defend, and the only one.

What does not improve. In the GATE Monte Carlo toolkit, the Mersenne Twister produced results statistically indistinguishable from quantum random number generators [28]. A 2025 linac study compared GATE's three PRNGs; the Mersenne Twister ran about 18% faster than James-Random [29]. Never claim better randomness gives better images.

What does improve: unbiased selection. A classical generator is deterministic: whoever picks the seed can try many seeds and keep the cohort that flatters their scanner ("seed shopping"), and nobody can detect it afterwards. The FDA names cohort-selection bias as a risk in virtual trials [3]. Quantum measurement outcomes are unpredictable even to the operator, and a certificate (section 5) proves it.

The hybrid design. Quantum draws only the start; a standard RNG grows the tumor (section 6). Seeding that RNG from the certified bits keeps the whole trajectory certified: nobody chose the start, and every later step is deterministic and replayable from the receipt.

What quantum cannot fix. If the published statistics under-represent a tumor type, every draw inherits it. That is rulebook bias, handled by citing sources and publishing the rulebook before generation.

5. Certifying the randomness

A CHSH value above 2, measured in the same hardware job that draws the tumors, proves the outcomes came from quantum measurement and not a prepared list. This is established physics with top-journal experiments behind every step.

What it proves	Paper	Venue
Bell violation certifies randomness	Pironio et al. [30]	Nature 2010
A small seed can be expanded into more certified randomness	Vazirani & Vidick [31]	STOC 2012
Expansion works in experiment (NIST, USTC)	Shalm et al. [32]; Liu et al. [33]	Nature Physics 2021
A commercial cloud quantum computer produces certified random bits, verified by a supercomputer	Liu et al. [34]	Nature 2025

What it proves	Paper	Venue
A public, traceable Bell-test randomness beacon (CURBy)	Kavuri et al. [35]	Nature 2025
Superconducting qubits (IBM's technology) pass a loophole-free Bell test	Storz et al. [36]	Nature 2023

Planned certificate layers.

1. **Hidden test rounds.** Every shot uses the same entangled qubits; a secret choice made after the circuit is fixed decides whether a shot is a CHSH test or a tumor draw. Passing tests certifies the draws, as in device-independent protocols [30, 31].
2. **Randomness expansion seeded by CURBy.** The loophole-free beacon [35] seeds the test-round choice; IBM expands it [32].
3. **Contextuality** (KCBS inequality) as a second, single-system certificate [37].
4. **Self-testing** from near-maximal CHSH values, certifying which state and measurements the device used [38].
5. **Hard-to-sample tumor shapes verified on PERUN** by cross-entropy benchmarking, as in [34, 39], with PERUN [61] in place of Frontier. Certified only against adversaries smaller than roughly PERUN scale.

Honest limit. A single IBM chip is not loophole-free; the certificate trusts the hardware. [36] shows the technology can close this; mixing in CURBy narrows the gap today.

6. Classical growth after the quantum seed

Once the quantum circuit fixes the starting tumor, a rule-based cellular automaton grows it on a classical computer, driven by a standard RNG seeded from the certified bits. Both the model type and its realism are already validated.

- **Kansal, Torquato et al. (Princeton)** is the classic cellular-automaton tumor model: from a few cells, proliferating, dormant and necrotic layers emerge from local rules [40].
- **Pixel2Cancer (Johns Hopkins)** grows tumors in real CT with three rules (growth, invasion, death); radiologists rated the results realistic in liver, pancreas and kidney [9].
- **Self-reproduction and evolution in cellular automata** reviews how far rule-based artificial life has come, useful background for growth and mutation rules [41].

Why this split. Growth needs millions of random draws, where classical generators are fast and quantum adds nothing measurable [28]. Only the start needs to be unchosen, so only the start is quantum.

Follow-up scans. Running the automaton longer gives a baseline and a later scan of the same tumor with known true growth, matching the prior-comparison feature of AI-Rad Companion Chest CT [54].

7. Inserting tumors into CT and simulating imaging

Start with image-domain insertion into public lung CT; Duke showed it matches projection-domain insertion within 3% in volume [42]. Lungs are the best start: high contrast against air, free annotated data and a direct Siemens product.

Approach	How	Evidence	When
Image domain	Draw the nodule into the reconstructed CT, blur to the scanner's sharpness, add matched noise	Recipe [11]; validated against real phantom nodules [42]	First version, any public scan
Projection domain	Add the nodule to raw scanner data before reconstruction	Mayo Clinic validation [43]	Upgrade; Siemens raw data available [44]
Full virtual patient	Phantom + scanner simulator	XCAT [6], DukeSim modelled on the Siemens NAEOTOM Alpha [46, 47], SYN-LUNGS [7]	Best fidelity; needs Duke access

Data. LIDC-IDRI provides about 1,000 annotated chest CTs [45]. The Mayo/TCIA library includes raw projection data from 50 low-dose chest screening scans on a Siemens scanner, free under CC BY 3.0 [44].

Realism checks. Compare inserted nodules with real annotated nodules of the same size (density profile, edge sharpness), run a radiologist reading test as in [9], and test a detector trained on the synthetic set against real cases, as in [8].

8. Provenance, signing and audit

Every tumor gets a signed receipt from which anyone can verify its origin and regenerate it exactly. This directly serves the reproducibility and credibility requirements set for virtual imaging trials [2].

The receipt holds: IBM job ID and calibration snapshot, raw counts, CHSH value, decoded tumor parameters, growth-model version and RNG seed, code hash and an Ed25519 signature, stored in an append-only ledger.

Commit before you draw. The rulebook and code hash are published to the ledger before the quantum job runs, so nobody can change the rules after seeing the tumors.

A third party can check: the signature; that CHSH exceeds 2; that the tumor regenerates from its parameters; and that the whole cohort matches the declared rulebook (chi-square or total-variation distance).

Why provenance matters now. CT-GAN injected and removed lung cancer in real CT scans convincingly enough to fool radiologists [48]. When synthetic and real images are indistinguishable, proof that an image is synthetic, and where it came from, becomes essential. Frame this as data traceability, not security.

9. Why it is useful

Synthetic tumors already train real detection AI and support regulatory decisions; this pipeline adds unchosen selection, patient-free generation and verifiable provenance.

Synthetic tumors train real AI. A model trained only on rule-based synthetic liver tumors segmented real tumors about as well as one trained on real data [8]; in further case studies, models trained with synthetic tumors matched or beat real-data training [10].

Regulators accept simulated evidence. VICTRE reproduced a real trial's result with 2,986 virtual patients [4]. Once virtual patients count as evidence, proving they were not cherry-picked matters [3].

No patient can leak. Latent diffusion models memorise training patients: one study found about 37% of patient data memorised and 69% of synthetic samples identified as copies [49]; diffusion models memorise more than GANs [50]. Learned quantum generators share the problem: a low training loss does not prove they generalise [51, 52]. A rule-based generator fed only published statistics sees no patient scan, so it cannot copy one.

Siemens fit. AI-Rad Companion Chest CT detects lung nodules and compares them with prior scans [54]; certified baseline-plus-follow-up pairs with known true growth are direct test data for it. DukeSim already models the Siemens NAEOTOM Alpha [46].

Where it wins: regulatory submissions using virtual trials, neutral cross-vendor detector benchmarks, shared synthetic datasets that must be provably synthetic, and longitudinal test sets. It does not win on image realism, speed or volume; classical tools are better there.

10. Why quantum, honest limits and hardware feasibility

Classical computers can generate the tumors; only quantum can prove nobody chose them. Everything below keeps that claim, and no larger one.

The strongest objection. "Commit-then-reveal plus a public random beacon blocks seed shopping classically." Partly true, and the pipeline uses commit-then-reveal anyway (section 8). Quantum adds randomness certified unpredictable even to its producer, with no trust in a beacon operator.

Benchmark like a sceptic. Large benchmarks show classical models often beat quantum ones, and removing entanglement often changes nothing [53]. The thesis therefore runs a classical baseline (PRNG + commit-then-reveal) on the same rulebook and reports what the quantum version adds and what it does not.

Hardware feasibility.

- IBM processors give reliable results at 127 qubits with error mitigation [55]; a 100-qubit virtual patient (context, four nodules, follow-ups, certificate pairs) is within reach.
- Noise will skew the cohort away from the rulebook; measuring that drift against patient size is a core, publishable result.
- If amplitude estimation is used to audit cohort statistics: the quadratic speedup is proven [56], but loading distributions can erase it [57], and on today's hardware the error plateaus at $(2–5)\times10^{-2}$ [58]. Full-scale advantage needs fault tolerance; finance needs about 7.5k logical qubits [59], radiation transport about 10^5 [60].
- PERUN (208 NVIDIA H200 GPUs, 10.7 PFlops) verifies about 36–40-qubit shape circuits by simulation [61].

What could fail, and the fallback. CHSH stuck below 2 → better qubit selection, error mitigation, or CURBy as the certified source. No Duke tools → image-domain insertion on public scans [11, 42]. Reviewers call quantum decorative → the measured drift curve and the reproducible pipeline still stand as results.

11. What is new

Each part is proven; the combination is not. In the searches behind this document, no published work combines these pieces (not found is not proven absent).

Part	Already done	Not yet done (this thesis)
Procedural lesions	Rule-based tumors and virtual trials [4, 7–9]	Rule-based cellular-automaton growth in **lungs**

Part	Already done	Not yet done (this thesis)
Quantum procedural generation	Game maps and images on IBM [12, 13]	Applied to **medical** content
Certified randomness	Bell, expansion, cloud QPU, beacon [30–36]	Used to certify a **virtual patient cohort**
Selection bias	Named as a risk by the FDA [3]	A mechanism that **proves** selection was unbiased
Provenance	Reproducibility required [2]	**Per-image** signed receipt with quantum certificate

Measurable results the thesis delivers: cohort drift versus patient size on IBM hardware; CHSH and contextuality values per job; verifier pass rate; a classical-baseline comparison; detector performance on the generated images.

Possible collaborators: Duke Center for Virtual Imaging Trials (Samei, Abadi, Segars) for phantoms and DukeSim; Johns Hopkins (Yuille, Zhou) for rule-based tumor synthesis; NIST and CU Boulder (Shalm, Knill) for CURBy; the FDA imaging division (Badano) for regulatory relevance.

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